

GeoMAMBA-AD: Early Detection of Alzheimer’s Disease Using Non-Invasive Resting-State fMRI and John Ellipsoid-Driven MAMBA Network

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Abstract

Early detection of Alzheimer’s disease (AD) is crucial for timely clinical treatment. Resting-state functional magnetic resonance imaging (rs-fMRI) can reveal early functional connectivity alterations that precede structural changes, but its use is constrained by limited data availability and scanner-induced domain shift. To address these challenges, we propose GeoMAMBA-AD, a framework that integrates John ellipsoid-based (JE) blind source separation (BSS) with a Mamba-based temporal learner to achieve robust cross-scanner generalization under small-data conditions. Exploiting a convex geometry formulation, JE can provably extract the perfect sources under rather mild data purity conditions, while avoiding the NP-hard computing and unreliable source independence assumptions used in ICA. Moreover, JE-BSS is performed at the individual-subject level, enabling source separation without enforcing shared component structures across subjects (e.g., group ICA). The resulting geometric embeddings are subsequently processed by a Mamba network, enabling stable and accurate AD classification across heterogeneous rs-fMRI datasets. GeoMAMBA-AD achieves 92% accuracy in cross-scanner scenarios under small-data learning settings, underscoring its robustness across different fMRI scanners and acquisition protocols. Furthermore, gradient-based interpretability analysis reveals that the geometry sources capture known executive control and subcortical control systems, both highly correlated to AD biomarkers, supporting the neurobiological plausibility of the JE sources. This is the first work that integrates geometry BSS into a Mamba backbone for early AD detection.

1. Introduction

Alzheimer’s disease (AD) is a progressive neurodegenerative disease and the most common cause of dementia. According to the World Health Organization (WHO), more than 55 million people currently live with dementia, and this

number is expected to nearly triple by 2050 due to global population aging [1]. AD is characterized by the gradual decline of cognitive functions, including memory, reasoning, and language, ultimately leading to severe functional impairment and loss of independence. Given the absence of curative treatments, early and accurate detection of AD is crucial. Detecting AD in its prodromal stages enables timely clinical interventions that can slow disease progression and improve patients’ quality of life, thereby reducing the socioeconomic burden on caregivers and healthcare systems. Recent therapeutic advances, such as the monoclonal antibody donanemab approved by the United States Food and Drug Administration in 2024 for early symptomatic AD, which targets amyloid pathology to slow clinical progression, further underscore the critical need for early diagnosis to maximize treatment efficacy [2]. Advanced AD research has associated the disease with several brain abnormalities, including amyloid- β (A β) plaques, hyperphosphorylated τ (p-tau) tangles, and myelin breakdown [3]. These changes often lead to the shrinkage of brain regions, notably the hippocampus, and altered neuronal communication within cortical networks.

Resting-state functional magnetic resonance imaging (rs-fMRI) is a prevalent neuroimaging technique used to detect these changes. It measures spontaneous blood-oxygen-level-dependent (BOLD) signal fluctuations throughout the brain without requiring task performance [4]. Studies have demonstrated positive correlations between AD-related cognitive decline and changes in BOLD signals measured by rs-fMRI [5–8]. Specifically, brain regions affected by neurodegeneration exhibit distinct BOLD signal patterns in AD patients compared to healthy individuals. This pathology-related difference makes rs-fMRI suitable for early AD detection. However, unlike structural MRI (sMRI), which has been extensively investigated for Alzheimer’s disease diagnosis [9–13], rs-fMRI introduces substantially greater algorithmic and methodological challenges due to its dynamic, temporally fluctuating neural signals. Specifically, its BOLD signals are indirect reflections of neural activity and are easily confounded by head motion,

physiological noise, and scanner drift, all of which introduce high variability across sessions and sites [14]. These factors make rs-fMRI datasets more difficult to harmonize and significantly less abundant than sMRI data, thereby limiting the scalability of deep learning approaches [15, 16]. Despite these challenges, the potential of rs-fMRI for early AD detection motivates us to develop algorithms that can overcome its limitations and fully leverage its functional information. Recent advances in AI modeling have introduced efficient sequence architectures such as Mamba [17–19], which have demonstrated strong performance in sMRI applications (e.g., Vision Mamba and AD-Diff [20, 21]); however, the potential of Mamba for fMRI remains largely unexplored, which motivates us to adopt Mamba as the backbone of our framework.

Real-world clinical datasets often consist of heterogeneous rs-fMRI scans acquired from different MRI scanners, making it difficult for deep neural networks to capture disease-specific features consistently [22, 23]. As a result, scanner-dependent biases and latent domain discrepancies frequently lead to cross-scanner overfitting, undermining model generalization. To address this challenge, we introduce a John ellipsoid-based blind source separation [24]. In convex geometry, the John ellipsoid defines an affine-invariant mapping that normalizes each domain’s feature distribution. Such normalization, when combined with individual-subject BSS, mitigates scanner-dependent variability and improves cross-scanner consistency in rs-fMRI analysis. This mapping minimizes domain-specific anisotropy and aligns heterogeneous scanners’ feature spaces, yielding a unified representation well-suited for downstream modeling. Meanwhile, the blind source separation step produces corresponding temporal dynamic curves (TDCs) that are subsequently processed through fully connected layers to encode the temporal information of rs-fMRI signals, while the spatial sources are fed into the Mamba network for feature extraction. This dual-path design enables the framework to capture spatial and temporal dynamics independently, obtaining the complementary information more effectively under multi-scanner and small-data conditions. In the next section, we introduce several seminal methods that have been developed for processing rs-fMRI data.

2. Related Work

2.1. Resting-State fMRI Analysis for AD

Traditional approaches typically rely on independent component analysis (ICA) [25] or related correlation-based measures to characterize brain connectivity alterations [26]. More recently, deep learning-based models such as 3D convolutional neural networks (3D-CNNs), recurrent neural networks (RNNs), and long short-term memory network

have been explored for automated AD classification using rs-fMRI data [9, 10, 27]. In particular, the brain connectivity graph convolutional network adopts a graph convolutional network (GCN) to encode inter-regional brain connectivity based on rs-fMRI signals [28]. Specifically, the method employs the anatomical automatic labeling (AAL) atlas to define 90 regions of interest (ROIs) and constructs subject-specific adjacency matrices by computing pairwise functional connectivity (i.e., Pearson correlation) between ROI time series. This design allows the GCN to exploit both node-level features and topological priors encoded in the graph structure, providing a relational inductive bias that can enhance generalization in limited-data settings. Spatiotemporal graph transformer extends this design by simultaneously incorporating both node-level activation features and inter-regional connectivity [29]. Specifically, it utilizes the anatomical automatic labeling (AAL) template to extract 116 regions of interest (ROIs) and constructs subject-wise relation matrices based on ROI correlations. These relation matrices are subsequently used to initialize the attention weights within a spatiotemporal transformer block, allowing the model to encode prior structural dependencies directly into the attention mechanism and jointly learn spatial-temporal representations from rs-fMRI data. Another representative approach, gICA 3D-VGG [30], combines group-wise ICA (gICA) with a 3D-CNN to extract shared functional components across subjects and perform volumetric classification. However, these models could require large-scale datasets and tend to exhibit limited generalization across scanners due to signal variability, scanner drift, and demographic heterogeneity.

2.1.1. Blind Source Separation and Convex Geometry in Neuroimaging

Blind source separation (BSS) has its roots in hyperspectral imaging and remote sensing, where geometry-based formulations have been developed to decompose mixed spectral signals into constituent material signatures without prior distributional assumptions [24, 31–34]. Unlike ICA, geometry-based BSS methods emphasize the convex and spatial separability of sources and therefore do not rely on statistical independence assumptions. When applied to rs-fMRI, this property allows geometry-based BSS to separate spatially overlapping but functionally related brain sources that would otherwise be confounded by ICA. Recent convex-geometry formulations further enhance BSS by introducing the John ellipsoid (JE), defined as the maximum-volume ellipsoid inscribed within a data convex hull, serving as an affine-invariant preconditioning scheme. In this work, JE-BSS is performed at the individual subject level, enabling source extraction without requiring cross-subject alignment or group-level factorization, which is particularly important when dealing with heterogeneous clinical cohorts.

2.2. Mamba and Selective State-Space Model

Learning the spatial representations of functional brain sources is essential for identifying disease-related alterations in rs-fMRI data. While transformer-based architectures have shown strong representational power for capturing spatial dependencies, their quadratic complexity limits scalability for small and high-dimensional fMRI datasets [29]. The recently proposed Mamba architecture, based on a selective state-space model (SSM), enables efficient long-range dependency learning with linear computational complexity [17]. Building upon these insights, we extend Mamba to functional imaging by coupling it with a JE-BSS framework, described in detail in Section 3.

3. Method

3.1. Method Overview

Conventional deep learning models trained on rs-fMRI often overfit to scanner-specific acquisition statistics under limited sample sizes, leading to shortcut learning that exploits scanner-dependent cues rather than disease-relevant neural patterns. To mitigate this limitation, we propose GeoMAMBA-AD, a hybrid framework that integrates a non-trainable John ellipsoid-based blind source separation (JE-BSS) stage with a Mamba-based spatiotemporal learning network. The JE-BSS module serves as a geometry-driven convex preconditioning stage, enforcing scanner-independent source separation through John ellipsoid preconditioning and simplex-constrained optimization. The spatial sources are then fed into the Mamba branch for feature learning, while their associated TDCs, estimated via convex optimization (CVX), are processed by fully connected layers to encode temporal dependencies.

An overview of the proposed GeoMAMBA-AD pipeline is shown in Fig. 1. We describe each step in detail as follows, covering both the non-trainable BSS stage and the trainable Mamba stage.

3.2. John Ellipsoid-Based Blind Source Separation

Let $\mathbf{Y}_1, \dots, \mathbf{Y}_L \in \mathbb{R}^n$ denote the rs-fMRI data matrix from one subject, with $n \approx 1\text{E}6$ is the number of voxels and L is the number of time points. The BSS factorization is formulated as

$$\mathbf{Y} \approx \mathbf{M}\mathbf{X} \in \mathbb{R}^{n \times L}, \quad (1)$$

where $\mathbf{Y} \triangleq [\mathbf{y}_1, \dots, \mathbf{y}_L]$ is the data matrix ($\mathbf{y}_l$: vectorization of $\mathbf{Y}_l$), $\mathbf{M} \triangleq [\mathbf{m}_1, \dots, \mathbf{m}_p] \in \mathbb{R}^{n \times p}$ is the source matrix ($\mathbf{m}_p$: vectorization of $\mathbf{M}_p$), and $\mathbf{X} \triangleq [\mathbf{x}_1, \dots, \mathbf{x}_p]^T \in \mathbb{R}^{p \times L}$ is the mixing matrix with $\mathbf{x}_p \in \mathbb{R}^L$ contains their temporal activation profiles (cf. the upper part of Fig. 1). Motivated by recent findings that rs-fMRI temporal variability can be largely explained by a small number of recurrent states [35, 36], we accordingly set $p = 4$ to obtain the basis representation.

Since the rs-fMRI data are extremely high-dimensional, we directly perform dimension reduction (DR) to an $(L-1)$ -dimensional subspace using an orthonormal temporal basis constructed via Gram–Schmidt, after column-wise centering each data point and projecting it onto this basis (detailed in the supplementary material). After DR, we construct a $(p-1)$ -dimensional affine transformation [37] and project all voxel trajectories into this reduced subspace, forming the vertex representation of a convex polytope (detailed in the supplementary material). Owing to the geometry-preserving property of DR and the ability of the John ellipsoid formulation to handle ill-conditioned data, this reduced representation remains suitable for the geometry blind source separation. Following John ellipsoid theory, we estimate the maximum-volume inscribed ellipsoid [24, Sec. III-A], yielding a preconditioned representation

$$\mathbf{Y}_{\text{pre}} = \mathbf{F}^*(\mathbf{Y} - \mathbf{c}^* \mathbf{1}_L^T), \quad (2)$$

where $\mathbf{F}^*$ and $\mathbf{c}^*$ denote the optimal parameters of the ellipsoid $E(\mathbf{F}, \mathbf{c})$, and $\mathbf{1}_L \in \mathbb{R}^{L \times 1}$ is a column vector of ones. The sources in this preconditioned space are then estimated by solving

$$\begin{aligned} \min_{\mathbf{M}_{\text{pre}} \in \mathbb{R}^{n \times p}, \mathbf{X} \in \mathbb{R}^{p \times L}} \quad & \|\mathbf{Y}_{\text{pre}} - \mathbf{M}_{\text{pre}}\mathbf{X}\|_F^2 \\ \text{s.t.} \quad & \mathbf{X} \geq 0, \quad \mathbf{1}_p^T \mathbf{X} = \mathbf{1}_L^T. \end{aligned} \quad (3)$$

where the simplex constraint enforces normalized nonnegative temporal profiles. Initialization uses the successive projections algorithm, and $\mathbf{X}$ is solved using CVX. To regularize $\mathbf{M}_{\text{pre}}$ toward a canonical simplex geometry, we further apply an orthogonal Procrustes update (1-step closed form via SVD). After estimation in the preconditioned space, the recovered sources are mapped back to the original scanner domain through the inverse affine transform defined by the John ellipsoid mapping [24, Sec. III-B]:

$$\widehat{\mathbf{M}} = (\mathbf{F}^*)^{-1} \mathbf{M}_{\text{pre}} + \mathbf{c}^* \mathbf{1}_L^T. \quad (4)$$

In this way, each column of $\widehat{\mathbf{M}}$ is treated as a spatial latent source signature, and each row of $\mathbf{X}$ is a temporal dynamic curve (TDC). These serve as inputs to the spatial and temporal branches of our Mamba architecture, respectively.

3.3. Mamba-Based Spatiotemporal Learning

Let p denote the number of latent sources estimated from the JE-BSS stage, L the number of time samples, and (H, W, D) the spatial resolution of each source map. The JE-BSS stage provides two outputs per subject: A set of spatial source maps $\widehat{\mathbf{M}} = \{\widehat{\mathbf{M}}_1, \widehat{\mathbf{M}}_2, \dots, \widehat{\mathbf{M}}_p\}$ with $\widehat{\mathbf{M}}_i \in \mathbb{R}^{H \times W \times D}$, and their TDCs $\mathbf{X} = \{\mathbf{x}_i\}_{i=1}^p$, where $\mathbf{x}_i \in \mathbb{R}^L$. These JE-BSS outputs provide voxel-level latent sources and their time-varying activation profiles, which are used as inputs to the Mamba-based spatiotemporal learning network for AD classification.

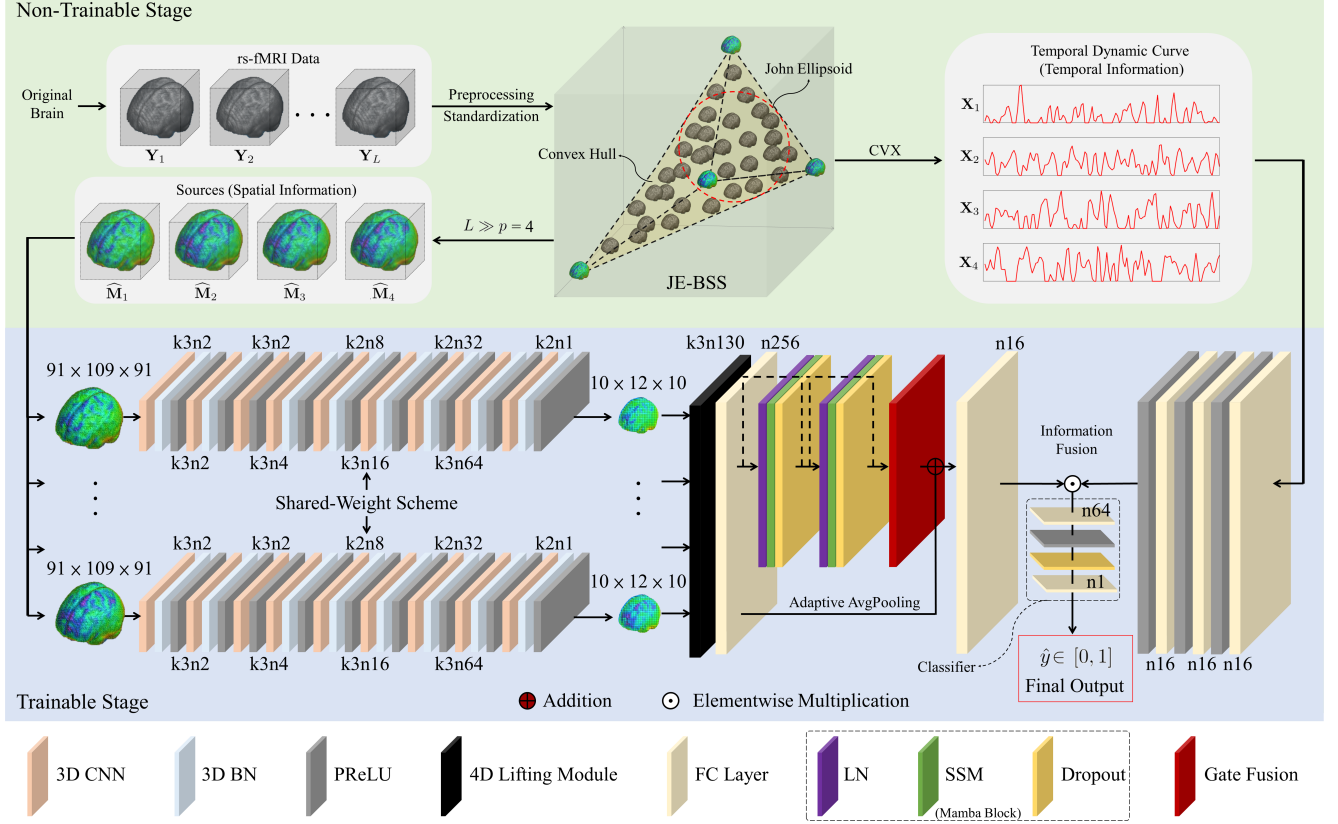


Figure 1. Overall architecture of the proposed GeoMAMBA-AD framework for AD early detection using rs-fMRI. The pipeline consists of a non-trainable JE-BSS stage and a trainable Mamba-based spatiotemporal learning stage. The JE-BSS stage performs source separation, producing spatial sources $\hat{M}$ and their corresponding TDCs X . In the trainable stage, spatial source maps are first compressed by a shared 3D CNN and lifted into temporal channels by a 4D lifting module, then flattened and projected before being processed by stacked Mamba blocks for temporal feature learning. A learnable adaptive pooling branch is fused with Mamba embeddings via element-wise multiplicative gating, where source-conditioned TDC features modulate the temporal representation. The fused representation is passed through fully connected layers to obtain a final AD probability.

Shared 3D CNN and Channel-to-Time Lifting. Each $\hat{M}_i$ is processed by a shared 3D convolutional compressor $g(\cdot)$ to extract latent spatial features, which can be written as

$$Z_i = g(\hat{M}_i) \in \mathbb{R}^{H' \times W' \times D'}. \quad (5)$$

The p feature maps are then concatenated along the channel axis and transformed by a 4D lifting module $\phi(\cdot)$ that maps source channels into temporal channels, which can be described as

$$\tilde{Z} = \phi([Z_1, Z_2, \dots, Z_p]) \in \mathbb{R}^{L \times H' \times W' \times D'}. \quad (6)$$

This aligns spatial features into a time-indexed representation.

Spatial Flattening and Temporal Sequence Construction. The lifted tensor $\tilde{Z}$ is reshaped into a time-sequence

matrix

$$Z_{\text{seq}} = [z_1, z_2, \dots, z_L]^T \in \mathbb{R}^{L \times P}, P = H'W'D', \quad (7)$$

and projected into an embedding space of dimension d , i.e.,

$$H_0 = Z_{\text{seq}} W_{\text{in}} \in \mathbb{R}^{L \times d}, \quad (8)$$

where W_{in} is a learnable linear projection matrix that maps each flattened spatial vector into a d -dimensional latent embedding space; each row of H_0 therefore represents the latent spatial configuration at a single fMRI time frame.

Temporal Compression via Mamba. To capture long-range temporal dependencies, we apply a selective state-space model (SSM) implemented by Mamba [17]. We augment H_0 with L learnable sink tokens $E \in \mathbb{R}^{L \times d}$ to form a vertically concatenated sequence

$$H_{\text{ext}} = \begin{bmatrix} H_0 \\ E \end{bmatrix} \in \mathbb{R}^{(2L) \times d}. \quad (9)$$

Let $\mathbf{H}^{(0)} = \mathbf{H}_{\text{ext}}$. We then apply L_{blk} stacked Mamba blocks, where $\text{Mamba}(\cdot)$ denotes a single Mamba block (state-space transition with input-dependent gating)

$$\mathbf{H}^{(\ell+1)} = \mathbf{H}^{(\ell)} + \text{Mamba}(\text{LN}(\mathbf{H}^{(\ell)})), \ell = 0, \dots, L_{\text{blk}} - 1. \quad (10)$$

After the Mamba blocks, the last token positions are extracted as compressed temporal embeddings $\mathbf{H}_{\text{tok}} \in \mathbb{R}^{L \times d}$. In parallel, an adaptive average pooling module generates $\mathbf{H}_{\text{base}} \in \mathbb{R}^{L \times d}$. The two representations are then fused through a learnable gate α , producing the final embedding

$$\mathbf{H}_{\text{fuse}} = \mathbf{H}_{\text{base}} + \tanh(\alpha) \mathbf{H}_{\text{tok}}, \quad (11)$$

where $\tanh(\alpha)$ adaptively balances the contributions of the baseline and token embeddings. This results in compact temporal embeddings that summarize dynamic transitions in vascular-mediated BOLD coupling.

Projection to Source-Interaction Space. The fused temporal embedding is projected into a K -dimensional semantic space, i.e.,

$$\mathbf{Y}_{\text{m}} = \mathbf{H}_{\text{fuse}} \mathbf{W}_{\text{out}} + \mathbf{B}_{\text{out}}, \quad \mathbf{Y}_{\text{m}} \in \mathbb{R}^{L \times K}, \quad (12)$$

where $\mathbf{W}_{\text{out}} \in \mathbb{R}^{d \times K}$ and $\mathbf{B}_{\text{out}} \in \mathbb{R}^K$ are trainable parameters, and $K = 16$ defines a compact latent space used to represent temporal features at each fMRI frame. Although 16 is numerically equal to $4p$ in our setting, this space does not enforce a one-to-one grouping of four dimensions per source; rather, it serves as a learned semantic basis that captures mixed-source interactions and AD-relevant temporal feature structure in a more expressive latent manifold.

Temporal Conditioning Using TDCs. We construct a time-indexed TDC matrix as

$$\mathbf{X} = [\mathbf{x}(1), \mathbf{x}(2), \dots, \mathbf{x}(L)] \in \mathbb{R}^{p \times L}. \quad (13)$$

Each $\mathbf{a}(t)$ represents the source-activation vector at time t , which is passed through an MLP $f_{\text{tdc}}(\cdot)$ to generate multiplicative temporal conditioning, i.e.,

$$\mathbf{Y}_{\text{x}}(\ell) = f_{\text{tdc}}(\mathbf{x}(\ell)) \in \mathbb{R}^K, \quad \ell = 1, \dots, L. \quad (14)$$

Collecting all time frames yields $\mathbf{Y}_{\text{a}} \in \mathbb{R}^{L \times K}$, which serves as data-driven gating to modulate the Mamba-derived temporal embeddings.

Fusion and Classification. Element-wise fusion integrates both components, written as

$$\mathbf{A} = \mathbf{Y}_{\text{m}} \odot \mathbf{Y}_{\text{a}} \in \mathbb{R}^{L \times K}. \quad (15)$$

The Hadamard product imposes multiplicative gating, enabling $\mathbf{Y}_{\text{a}}$ to selectively amplify or suppress entries of

$\mathbf{Y}_{\text{m}}$, thereby controlling interaction features and shaping gradient flow during training. Such multiplicative fusion is aligned with input-conditioned gating mechanisms that have been observed to influence representation learning and gradient routing during training [38].

A shallow classifier comprising several fully connected layers then produces the final prediction

$$\hat{y} = \text{Softmax}(f_2(\sigma(f_1(\text{vec}(\mathbf{A}))))). \quad (16)$$

This integrates the temporally evolving gated interaction features into a final AD decision.

4. Experimental Results

We present quantitative results for GeoMAMBA-AD in Sec. 4.3, followed by ablation studies on the BSS stage and track-level contributions (Sec. 4.4 and Sec. 4.5). Finally, we perform integrated gradients-based interpretability analysis to identify AD-related functional networks from the geometry sources in Sec. 4.6.

4.1. ADNI

Data used in the present study was obtained from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database (adni.loni.usc.edu). The ADNI was launched in 2003 as a public-private partnership, led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to test whether serial magnetic resonance imaging (MRI), positron emission tomography (PET), other biological markers, and clinical and neuropsychological assessment can be combined to measure the progression of MCI and early AD.

4.2. Datasets and Experimental Setup

We used resting-state fMRI (rs-fMRI) and T1-weighted anatomical images from the Alzheimer’s Disease Neuroimaging Initiative (ADNI) database. Two subsets were included: scans acquired on 3.0-Tesla Philips scanners and on 3.0-Tesla Siemens MAGNETOM Prisma-fit scanners.

For the Philips subset, rs-fMRI data were collected with echo-planar imaging (EPI) and a voxel resolution of $3 \times 3 \times 3$ mm³ over 140 time points. The Siemens subset used gradient-echo EPI with a voxel resolution of $3.4 \times 3.4 \times 3.4$ mm³ and 197 volumes per session. T1-weighted anatomical images (3D gradient-echo for Philips, MPRAGE for Siemens) were included for standard preprocessing (segmentation, normalization, and coregistration). In total, 100 subjects were used, including both Alzheimer’s disease (AD) and normal control (NC) participants. We first held out 25 subjects as the test set. The remaining 75 subjects were split into training and validation sets using a random 80/20 split.

The model was trained with a batch size of 4 and optimized using the AdamW optimizer with a learning rate of

396 3×10^{-4} and weight decay of 10^{-4} . Early stopping was ap- 448
 397 plied based on the validation κ . Training was terminated if 449
 398 κ did not improve for 10 consecutive epochs, and the check- 450
 399 point with the highest validation κ was used for final testing. 451
 400 We compare with transformer-based, graph convolutional 452
 401 (GCN), and 3D CNN baselines. Implementation details are 453
 402 provided in the supplementary material. 454

403 4.3. Quantitative Results 455

404 We evaluate GeoMAMBA-AD under a realistic clinical de- 456
 405 ployment scenario characterized by scanner-induced do- 457
 406 main shift. The definitions of all evaluation metrics are 458
 407 provided in the supplementary material. This experiment 459
 408 examines whether the model can learn disease-relevant tem- 460
 409 poral representations rather than relying on scanner-specific 461
 410 statistics, thereby providing a more faithful measure of clin- 462
 411 ical generalizability. This setup reflects a common chal- 463
 412 lenge in neuroimaging research, where increasing sample 464
 413 size across clinical centers inevitably introduces domain 465
 414 shifts due to scanner-specific hardware and acquisition dis- 466
 415 crepancies. To simulate this, we allocate 75 subjects for 467
 416 training validation and designate 25 subjects (all randomly 468
 417 sampled from each scanner) as a cross-scanner dataset. In 469
 418 practice, this mixed-domain training often leads to the op- 470
 419 posite effect: instead of learning stable disease signatures, 471
 420 the network rapidly locks onto scanner-specific low-level 472
 421 statistics as the most discriminative cues during optimiza- 473
 422 tion. Since scanner-induced intensity distributions are often 474
 423 more dominant and easier to separate than subtle pathologi- 475
 424 cal patterns, the model prioritizes domain separation over 476
 425 disease separation, resulting in a latent space that is well- 477
 426 clustered by scanner, but not really aligned with clinical 478
 427 labels. With John ellipsoid-based source separation, the 479
 428 geometry-stabilized latent sources become less sensitive to 480
 429 cross-scanner domain shift, enabling GeoMAMBA-AD to 481
 430 achieve SOTA performance across scanners. 482

431 Compared to this, several peer methods reveal varying 483
 432 degrees of sensitivity to domain shifts. BCGCN, which em- 484
 433 ploys graph convolutional networks to model brain data us- 485
 434 ing only ROI-based features, underperforms in this setting. 486
 435 Its lack of temporal information matrices limits its abil- 487
 436 ity to capture individualized dynamic changes of network 488
 437 structures, thereby reducing generalizability across hetero- 489
 438 geneous domains. STGTN incorporates a spatiotemporal 490
 439 transformer architecture with adjacency initialization (via 491
 440 AAL’s 116-ROI template), graph modeling, and adversarial 492
 441 training. Among all peer methods, it demonstrates the most 493
 442 stable performance under cross-scanner conditions. This 494
 443 suggests that while its architectural components help mit- 495
 444 igate scanner-induced variability, substantial domain shifts 496
 445 still pose challenges even for robust models. Finally, we 497
 446 replicate the gICA 3D-VGG model, which contains over
 447 86 million parameters, to evaluate the generalizability of

group ICA-based approaches. While group ICA is theo-
 retically designed to extract common functional networks
 across subjects by jointly decomposing the entire dataset, its
 effectiveness can be compromised under cross-scanner sce-
 narios. Scanner heterogeneity in signal characteristics may
 distort the group-level decomposition, resulting in compo-
 nents that fail to capture scanner-independent neural pat-
 terns. In our experiments, the gICA 3D-VGG model per-
 forms suboptimally on the cross-scanner dataset, suggesting
 that the representations learned through ICA do not gener-
 alize well across imaging domains—despite their intended
 inter-subject consistency. In contrast, our method (based on
 JE-based BSS and a dual-track structure) achieves higher
 accuracy with only 1976,260 parameters, nearly 4 times
 fewer than the reproduced VGG model. These results un-
 derSCORE the benefit of combining a source separation ap-
 proach that relaxes the independence assumption with dual-
 track feature learning, enabling robust performance under
 scanner-induced heterogeneity.

467 4.4. Ablation of Non-trainable Stage 468

469 In this experiment, we evaluate the robustness of differ- 470
 471 ent blind source decomposition strategies used in the non- 472
 473 trainable stage of the framework. Specifically, we com- 474
 475 pare the widely-used group-wise Independent Component 476
 477 Analysis (gICA) with our proposed geometric ellipsoid- 478
 479 based blind source separation (JE-BSS). To ensure fair 479
 480 comparison, the entire downstream trainable pipeline of 480
 481 GeoMAMBA-AD is kept identical across both settings; 481
 482 only the non-trainable decomposition module is switched. 482

483 As shown in Table 2, replacing JE-BSS with gICA leads 484
 485 to a substantial performance drop (OA: 92.00% down to 485
 486 76.00%, κ : 83.87 down to 48.28), confirming that conven- 486
 487 tional ICA-based latent source extraction is less stable in 487
 488 heterogeneous rs-fMRI scenarios. When comparing against 488
 489 peer methods that also rely on gICA-based preprocessing 489
 490 (cf. Table 1), we observe that even with the same latent 490
 491 source initialization, using a lightweight Mamba backbone 491
 492 together with our temporal interaction modeling (cf. Sec- 492
 493 tion 4.5 for temporal branch analysis) still yields perfor- 493
 494 mance gains (higher precision). This suggests that Mamba 494
 495 is naturally suitable for scenarios with limited training sam- 495
 496 ples. Although most prior Mamba studies have been con- 496
 497 ducted outside the neuroimaging domain, recent work has 497
 already shown that Mamba-based models can outperform
 CNN architectures under data-scarce medical settings [39].
 Motivated by this, we investigate whether such small-data
 advantages of Mamba can also transfer to rs-fMRI based
 Alzheimer’s disease classification. Our results show that
 this advantage carries over to rs-fMRI, even when the up-
 stream latent decomposition relies on gICA.

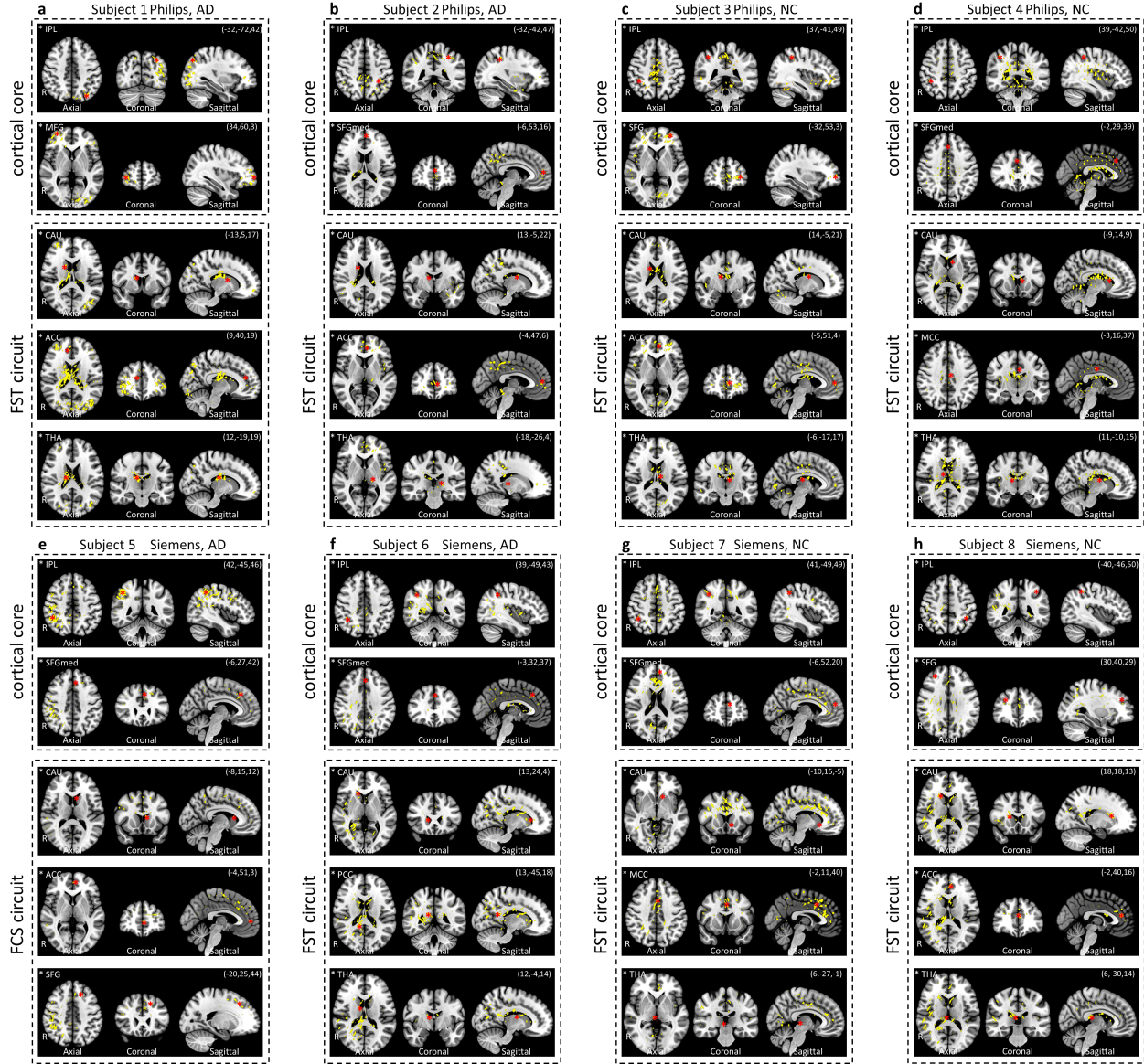


Figure 2. IG-based ROI attribution visualization of the first geometry source for 8 selected subjects across Philips and Siemens scanners under both AD and NC conditions. Plots (a–d) correspond to Philips subjects (AD: a–b; NC: c–d), and plots (e–h) correspond to Siemens subjects (AD: e–f; NC: g–h). Peak IG values highlight three control-related subsystems: the cortical Frontoparietal Control Network (FPN) core (MFG, SFG/SFGmed, IPL), the Fronto–Cingulo–Striatal (FCS) circuit (ACC, CAU, SFG), and the canonical Fronto–Striatal–Thalamo (FST) recurrent loop (ACC, CAU, THA).

4.5. Ablation of Spatiotemporal Stage

To further examine the contribution of each trainable component, we conduct ablation experiments on the trainable stage of GeoMAMBA-AD. In this ablation study, we isolate the Mamba-based branch and the temporal conditioning branch, evaluating each individually while keeping the non-trainable JE-BSS stage fixed (cf. Table 3).

This ablation is necessary to confirm that the great performance does not originate from a single dominant com-

ponent, but rather from the synergy of both trainable modules. As shown in Table 3, neither branch alone is sufficient to achieve satisfactory performance under limited rs-fMRI training data. Using the Mamba branch alone yields 68.00% OA, while relying solely on the temporal interaction branch further degrades performance. In contrast, when both mechanisms are jointly enabled, GeoMAMBA-AD reaches 92.00% OA and 83.87 κ , demonstrating that the two branches are complementary rather than redundant.

Table 1. Comparison of different deep learning methods. Bold-faced entries denote the highest performance scores.

Method	OA (%)	$\kappa(\times 100)$	Precision	Recall	F1 score
BCGCN [28]	68.00	28.57	0.67	0.40	0.50
gICA 3D-VGG [30]	72.00	40.67	0.67	0.60	0.63
STGTN [29]	72.00	46.15	0.60	0.90	0.72
GeoMAMBA-AD	92.00	83.87	0.83	1.00	0.91

Table 2. Ablation of the non-trainable stage. Bold-faced entries denote the highest performance scores.

Method	OA (%)	$\kappa(\times 100)$	Precision	Recall	F1 score
gICA	76.00	48.28	0.75	0.60	0.67
JE-BSS	92.00	83.87	0.83	1.00	0.91

Table 3. Ablation of the trainable stage. Bold-faced entries denote the highest performance scores.

Method	OA (%)	$\kappa(\times 100)$	Precision	Recall	F1 score
Mamba branch only	68.00	31.03	0.62	0.50	0.56
Temporal branch only	64.00	21.05	0.57	0.40	0.47
GeoMAMBA-AD	92.00	83.87	0.83	1.00	0.91

This indicates that both selective state-space feature learning and dynamic temporal information are necessary for stabilizing downstream learning after JE-BSS decomposition, and their synergy is critical for extracting reliable AD-discriminative representations from rs-fMRI. Importantly, this also aligns with clinical experience, where temporal dynamics carry meaningful physiological information [40]; yet peer gICA-based pipelines tend to discard the corresponding matrices entirely [30], which could partially explain their inferior performance under this setting.

4.6. Gradient-Based Identification of Neural Control Subsystems

Unlike group-wise statistical decomposition methods that directly project shared group-level spatial patterns, our individual geometry BSS extracts subject-specific latent sources. Therefore, it is necessary to examine whether these individualized sources still reflect known neurobiological networks. In this study, we specifically used peak integrated gradients (IG) attribution values [41] to quantify which anatomical ROIs contribute most strongly, ensuring that only the most discriminative nodes were considered (detailed implementation is provided in supplementary material). To this end, we applied IG attribution analysis to identify which anatomical ROIs the AI model relied on when making the AD/NC decision based on the geometry-derived source representations. This enables direct inspection of where the model assigns biological evidence during inference, as demonstrated in Fig. 2, where only the first source of each subject is visualized.

IG attribution maps from the first source of each subject consistently revealed three functional control subsystems,

each exerting distinct influence on the model’s AD classification decision:

- **Frontoparietal Control Network (FPN) Core:** High IG contributions were observed in the middle frontal gyrus (MFG), superior frontal gyrus (SFG/SFGmed), and inferior parietal lobule (IPL), as Fig. 2 shown. These cortical regions form the canonical FPN core supporting executive control. Their strong contributions suggest that disruption of top-down control processes provides salient discriminative cues for AD [42, 43].
- **Fronto-Cingulo-Striatal (FCS) Control Circuit:** A second subsystem involved the anterior cingulate cortex (ACC), caudate nucleus (CAU), and superior frontal gyrus (SFG), as illustrated in Fig. 2e. This FCS circuit corresponds to a cortico-cingulo-striatal executive control pathway, which has been linked to frontal-striatal contributions to cognitive control and behavioral regulation [44, 45].
- **Fronto-Striatal-Thalamo (FST) Recurrent Circuit:** A third cluster included the ACC, CAU, and thalamus (THA). This FST circuit corresponds to the canonical cortico-basal-ganglia-thalamic recurrent loop (cf. Fig. 2). The contribution of this recurrent control loop suggests that disruption of closed-loop executive regulation further supports AD vs. NC differentiation [46, 47].

Overall, these results demonstrate that the proposed geometry-based source extraction does not produce arbitrary spatial mixtures, but identifies three well-characterized executive control subsystems (FPN core, FCS, and FST) known to be impacted during early-stage Alzheimer’s disease. In addition, IG-based interpretability further shows that the model relies on these disease-relevant networks when making AD decisions, providing alignment with clinically recognized neurocognitive mechanisms and enhancing the trustworthiness of the proposed framework from a clinical interpretation perspective.

5. Conclusion

In this work, we presented GeoMAMBA-AD, a John ellipsoid-based blind source separation and Mamba-based spatiotemporal learning framework for Alzheimer’s disease detection from small-sample rs-fMRI. By introducing individual-level JE-BSS, selective state-space feature learning, and temporal conditioning, the proposed method achieves high disease separability (92% OA) and maintains robust discriminative performance under heterogeneous small-data and mixed-scanner conditions. Moreover, IG-based analysis identifies functional networks and control circuits (e.g., FPN, FCS, and FST) previously linked to AD, supporting the neurobiological relevance of the geometry sources. Overall, integrating geometric priors with Mamba offers a promising direction toward clinically reliable rs-fMRI early AD detection.

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